4. Power supply is switched on and gel is run till the tracer dye reaches the end of the gel.

As the DNA is negatively charged, DNA fragments move through the gel towards the positive electrode. The rate of migration of DNA is dependent on the size and shape. In general, smaller linear fragments move faster than the larger ones. Hence, gel electrophoresis can be conveniently used for the separation of a mixture of DNA fragments, based on their size. The bands of the DNA can be detected by soaking the gel in ethidium bromide solution (Ethidium bromide can also be added to molten agarose prior to setting the gel). When activated by ultraviolet radiation, DNA base pairs in association with ethidium bromide, emit orange fluorescence. And in this way the DNA fragments separated in agarose electrophoresis can be identified (Figure 12.30b).

PAGE is composed of chains of acryl amide monomers crosslinked with methylene bisacryalmide units. The pore size of the gel is dependent on the total concentration of monomers and the cross links. PAGE is used for the separation of single stranded DNA molecules that differ in length by just one nucleotide. Agarose gels cannot be used for this purpose. This is because polyacrylamide gels have smaller pore sizes than agarose gels and allow precise separation of DNA molecule from 10–1500 bp.

HOTS

1. Explain how gel electrophoresis can be used to determine the size of a PCR product.

12.11.2 Polymerase Chain Reaction (PCR)

The PCR technique has already proven exceptionally valuable in many areas of molecular biology, medicine, and biotechnology. PCR technique has great practical importance and impact on biotechnology. Between 1983 and 1985 American biochemist Kary Mullis developed PCR technique that made it possible to synthesize large quantities of a DNA fragment without cloning it. Mullis received the 1993 Nobel Prize for Chemistry for his invention. PCR is a cell free amplification technique.

Infobits

Cloned genes and other DNA sequences often are analyzed to determine the arrangement and specific locations of restriction sites. The analytical process involves cleavage of the DNA with restriction enzymes, followed by separation of the resulting DNA fragments by agarose gel electrophoresis. The sizes of the DNA fragments are calculated, enabling restriction maps to be constructed. The many DNA fragments produced by cleaving genomic DNA show a wide range of sizes, resulting in a continuous smear of DNA fragments in the gel. In this case, specific gene fragments can be visualized only by transferring them to membrane filter by southern blotting, hybridizing a specific labelled probe with the DNA fragments, and detecting the hybrids. A similar procedure, Northern blotting is used to analyze the sizes and quantities of RNAs isolated from cell.

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